KAIZEN MATHS

Lesson starter

Continuous Random Variables and Density Functions

Density & CDF

Name

S1

f(x) = kx for 0 ≤ x ≤ 4, and f(x) = 0 otherwise.Find k, then find P(0 < X < 3).

S2

f(x) = kx^2 for 0 ≤ x ≤ 2, and f(x) = 0 otherwise.Find k, then find P(0 < X < 1).

S3

The density of X is f(x) = $\frac{2}{25}x$ for 0 ≤ x ≤ 5, and 0 otherwise.Find F(x), then find P(X ≤ 1).

S4

For 0 ≤ x ≤ 6, F(x) = $\frac{x^2}{36}$.Find P(3 < X < 6).

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Continuous Random Variables and Density Functions

Density & CDF

Name _____

E1

f(x) = kx for 0 ≤ x ≤ 4, and f(x) = 0 otherwise.Find k, then find P(1 < X < 4).

E2

f(x) = kx^2 for 0 ≤ x ≤ 4, and f(x) = 0 otherwise.Find k, then find P(1 < X < 3).

E3

The density of X is f(x) = $\frac{2}{9}$x for 0 ≤ x ≤ 3, and 0 otherwise.Find F(x), then find P(X ≤ 1).

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Exit ticket

Continuous Random Variables and Density Functions

Density & CDF

Name _____

E1

f(x) = kx for 0 ≤ x ≤ 4, and f(x) = 0 otherwise.Find k, then find P(1 < X < 4).

E2

f(x) = kx^2 for 0 ≤ x ≤ 4, and f(x) = 0 otherwise.Find k, then find P(1 < X < 3).

E3

The density of X is f(x) = $\frac{2}{9}$x for 0 ≤ x ≤ 3, and 0 otherwise.Find F(x), then find P(X ≤ 1).

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Continuous Random Variables and Density Functions

Density & CDF

Name _____

E1

f(x) = kx for 0 ≤ x ≤ 4, and f(x) = 0 otherwise.Find k, then find P(1 < X < 4).

E2

f(x) = kx^2 for 0 ≤ x ≤ 4, and f(x) = 0 otherwise.Find k, then find P(1 < X < 3).

E3

The density of X is f(x) = $\frac{2}{9}$x for 0 ≤ x ≤ 3, and 0 otherwise.Find F(x), then find P(X ≤ 1).

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Continuous Random Variables and Density Functions

Density & CDF

Name _____

E1

f(x) = kx for 0 ≤ x ≤ 4, and f(x) = 0 otherwise.Find k, then find P(1 < X < 4).

E2

f(x) = kx^2 for 0 ≤ x ≤ 4, and f(x) = 0 otherwise.Find k, then find P(1 < X < 3).

E3

The density of X is f(x) = $\frac{2}{9}$x for 0 ≤ x ≤ 3, and 0 otherwise.Find F(x), then find P(X ≤ 1).

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Teacher answers & working

Continuous Random Variables and Density Functions · Density & CDF

STA-9F19-L1 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S1

f(x) = kx for 0 ≤ x ≤ 4, and f(x) = 0 otherwise.Find k, then find P(0 < X < 3).

Answer:

$k = \frac{1}{8}, P = \frac{9}{16}$

Working:

For a probability density function, the total area under the curve must be 1.

The support is 0 ≤ x ≤ 4, so set up ∫ from 0 to 4 of f(x) dx = 1.

Substitute f(x) = kx: ∫ from 0 to 4 of kx dx = 1.

Integrate kx. Since k is constant and ∫ x dx = $\frac{x^2}{2}$, the antiderivative is k$\frac{x^2}{2}$.

Evaluate the limits: [k$\frac{x^2}{2}$] from 0 to 4 = k$\frac{16}{2}$ - k$\frac{0^2}{2}$ = k$\frac{16}{2}$.

Solve k × $\frac{16}{2}$ = 1, so k = $\frac{2}{16}$ = $\frac{1}{8}$.

To find a probability, integrate the density between the two values.

P(0 < X < 3) = ∫ from 0 to 3 of ($\frac{1}{8}$)x dx.

The antiderivative of ($\frac{1}{8}$)x is $\frac{1}{16}x^2$.

Substitute the limits: [$\frac{1}{16}x^2$] from 0 to 3 = $\frac{(9-0)}{16}$.

$P = \frac{9}{16} \approx 0.563$

S2

f(x) = kx² for 0 ≤ x ≤ 2, and f(x) = 0 otherwise.Find k, then find P(0 < X < 1).

Answer:

$k = \frac{3}{8}, P = \frac{1}{8}$

Working:

Make the total area under the density equal to 1.

The support is 0 ≤ x ≤ 2, so ∫ from 0 to 2 of f(x) dx = 1.

Substitute f(x) = kx²: ∫ from 0 to 2 of kx² dx = 1.

Integrate kx². Since k is constant and ∫ x² dx = $\frac{x^3}{3}$, the antiderivative is k$\frac{x^3}{3}$.

Evaluate the limits: [k$\frac{x^3}{3}$] from 0 to 2 = k$\frac{8}{3}$ - k$\frac{0^3}{3}$ = k$\frac{8}{3}$.

Solve k × $\frac{8}{3}$ = 1, so k = $\frac{3}{8}$.

Now integrate the density across the requested interval.

P(0 < X < 1) = ∫ from 0 to 1 of ($\frac{3}{8}$)x² dx.

The antiderivative of ($\frac{3}{8}$)x² is $\frac{1}{8}x^3$.

Substitute the limits: [$\frac{1}{8}x^3$] from 0 to 1 = $\frac{(1-0)}{8}$.

$P = \frac{1}{8} \approx 0.125$

Teacher answers & working

Continuous Random Variables and Density Functions · Density & CDF

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S3

The density of X is f(x) = $\frac{2}{25}$x for 0 ≤ x ≤ 5, and 0 otherwise.Find F(x), then find P(X ≤ 1).

Answer:

$F(x) = \frac{x^2}{25}$ on $0 \leq x \leq 5$, $P = \frac{1}{25}$

Working:

The cumulative distribution function is F(x) = P(X ≤ x).

For 0 ≤ x ≤ 5, add up the density from the start of the support to x.

Use a dummy variable t so the upper limit can be x: F(x) = ∫ from 0 to x of ($\frac{2}{25}$)t dt.

Integrate ($\frac{2}{25}$)t. The antiderivative is $\frac{1}{25}t^2$.

Evaluate the limits: [$\frac{1}{25}t^2$] from 0 to x = $\frac{x^2}{25}$ - $\frac{0^2}{25}$ = $\frac{x^2}{25}$.

So $F(x) = 0$ for $x < 0$, $F(x) = \frac{x^2}{25}$ for $0 \leq x \leq 5$, and $F(x) = 1$ for $x > 5$.

$P(X \leq 1) = F(1) = \frac{1}{25}$.

$P = \frac{1}{25} \approx 0.040$

S4

For 0 ≤ x ≤ 6, F(x) = $\frac{x^2}{36}$.Find P(3 < X < 6).

Answer:

$\frac{3}{4}$

Working:

A CDF is already the integrated density, so use differences of F values.

For any continuous random variable, P(a < X < b) = F(b) - F(a).

$P(3 < X < 6) = F(6)$ - $F(3)$.

$F(6) = 1$ and $F(3) = \frac{1}{4}$.

$P = \frac{(36-9)}{36}$.

$P = \frac{3}{4} \approx 0.750$

Teacher answers & working

Continuous Random Variables and Density Functions · Density & CDF

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E1

f(x) = kx for 0 ≤ x ≤ 4, and f(x) = 0 otherwise.Find k, then find P(1 < X < 4).

Answer:

$k = \frac{1}{8}$, $P = \frac{15}{16}$

Working:

For a probability density function, the total area under the curve must be 1.

The support is 0 ≤ x ≤ 4, so set up ∫ from 0 to 4 of f(x) dx = 1.

Substitute f(x) = kx: ∫ from 0 to 4 of kx dx = 1.

Integrate kx. Since k is constant and ∫ x dx = $\frac{x^2}{2}$, the antiderivative is k$\frac{x^2}{2}$.

Evaluate the limits: [k$\frac{x^2}{2}$] from 0 to 4 = k$\frac{16}{2}$ - k$\frac{0^2}{2}$ = k$\frac{16}{2}$.

Solve k × $\frac{16}{2}$ = 1, so k = $\frac{2}{16}$ = $\frac{1}{8}$.

To find a probability, integrate the density between the two values.

P(1 < X < 4) = ∫ from 1 to 4 of ($\frac{1}{8}$)x dx.

The antiderivative of ($\frac{1}{8}$)x is $\frac{1}{16}x^2$.

Substitute the limits: [$\frac{1}{16}x^2$] from 1 to 4 = $\frac{(16-1)}{16}$.

$P = \frac{15}{16} \approx 0.938$

E2

f(x) = kx² for 0 ≤ x ≤ 4, and f(x) = 0 otherwise.Find k, then find P(1 < X < 3).

Answer:

$k = \frac{3}{64}$, $P = \frac{13}{32}$

Working:

Make the total area under the density equal to 1.

The support is 0 ≤ x ≤ 4, so ∫ from 0 to 4 of f(x) dx = 1.

Substitute f(x) = kx²: ∫ from 0 to 4 of kx² dx = 1.

Integrate kx². Since k is constant and ∫ x² dx = $\frac{x^3}{3}$, the antiderivative is k$\frac{x^3}{3}$.

Evaluate the limits: [k$\frac{x^3}{3}$] from 0 to 4 = k$\frac{64}{3}$ - k$\frac{0^3}{3}$ = k$\frac{64}{3}$.

Solve k × $\frac{64}{3}$ = 1, so k = $\frac{3}{64}$.

Now integrate the density across the requested interval.

P(1 < X < 3) = ∫ from 1 to 3 of ($\frac{3}{64}$)x² dx.

The antiderivative of ($\frac{3}{64}$)x² is $\frac{1}{64}x^3$.

Substitute the limits: [$\frac{1}{64}x^3$] from 1 to 3 = $\frac{(27-1)}{64}$.

$P = \frac{13}{32} \approx 0.406$

Teacher answers & working

Continuous Random Variables and Density Functions · Density & CDF

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E3

The density of X is f(x) = $\frac{2}{9}$x for 0 ≤ x ≤ 3, and 0 otherwise.Find F(x), then find P(X ≤ 1).

Answer:

$F(x) = \frac{x^2}{9}$ on $0 \leq x \leq 3$, $P = \frac{1}{9}$

Working:

The cumulative distribution function is F(x) = P(X ≤ x).

For 0 ≤ x ≤ 3, add up the density from the start of the support to x.

Use a dummy variable t so the upper limit can be x: F(x) = ∫ from 0 to x of ($\frac{2}{9}$)t dt.

Integrate ($\frac{2}{9}$)t. The antiderivative is $\frac{1}{9}t^2$.

Evaluate the limits: [$\frac{1}{9}t^2$] from 0 to x = $\frac{x^2}{9}$ - $\frac{0^2}{9}$ = $\frac{x^2}{9}$.

So $F(x) = 0$ for $x < 0$, $F(x) = \frac{x^2}{9}$ for $0 \leq x \leq 3$, and $F(x) = 1$ for $x > 3$.

$P(X \leq 1) = F(1) = \frac{1}{9}$.

$P = \frac{1}{9} \approx 0.111$